

PCL Technologies Inc.

眾達科技

(4977.TT)

2025.12.09

投資安全聲明 DISCLAIMER

本簡報所提供之資訊包含預測、預估或其他資訊之前瞻性聲明，因此具有風險及不確定因素，可能導致實際結果與前瞻性聲明中所表達的有重大差異。

This material has been prepared by PCL Technologies, Inc. (“PCL”).

Statements made in this material include forward-looking statements, which include, without limitation, statements about the issues, plans and expectations of PCL. Forward-looking statements reflect, among other things, management plans and objectives for future operations, current views with respect to future events and future economic performances and projections of various financial items. These forward-looking statements involve known and unknown risks, uncertainties, and other factors which may cause actual results to differ materially from those implied by such forward-looking statements.

公司介紹

眾達科技股份有限公司

* 成立日期: 2007.10.16

* 上市日期: 2013.11.20

* 資本額: 新台幣 8.02 億元

* 生產基地:

✓ 眾達光通科技(蘇州)有限公司

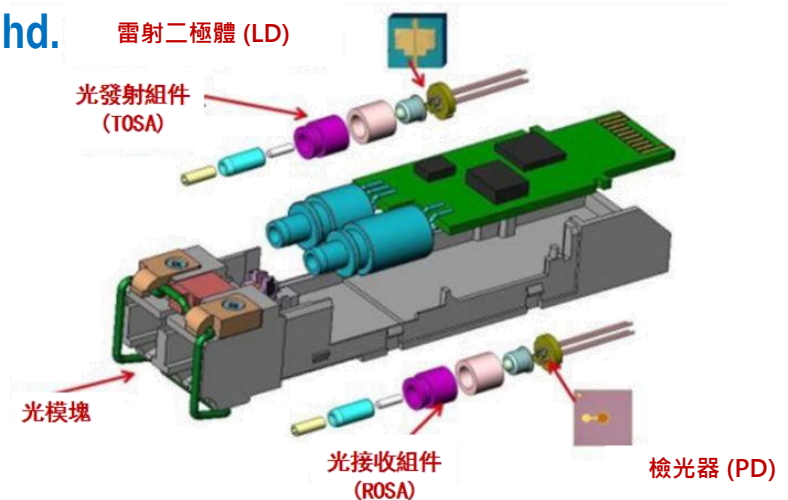
✓ PCL International Technologies (Penang) Sdn Bhd.

* 主要產品:

✓ 高速光收發模組

✓ 光發射/接收次模組

✓ 雷射及檢光器元件封裝



生產基地

PCL
TECHNOLOGIES

眾達光通科技(蘇州)有限公司



PCL International Technologies (Penang) Sdn Bhd. (檳城)





DATA CENTER MARKET (2025 - 2031)

Market size is estimated to reach
\$605 billion by 2031
CAGR of 10.3% during 2025 - 2031



Regional Analysis



BY INFRASTRUCTURE

- ❖ IT Infrastructure
- ❖ Electrical Infrastructure
- ❖ General Construction
- ❖ Support Infrastructure

BY OPERATOR

- ❖ Enterprise Data Centers
- ❖ Colocation Providers
- ❖ Cloud Service Providers (CSPs)

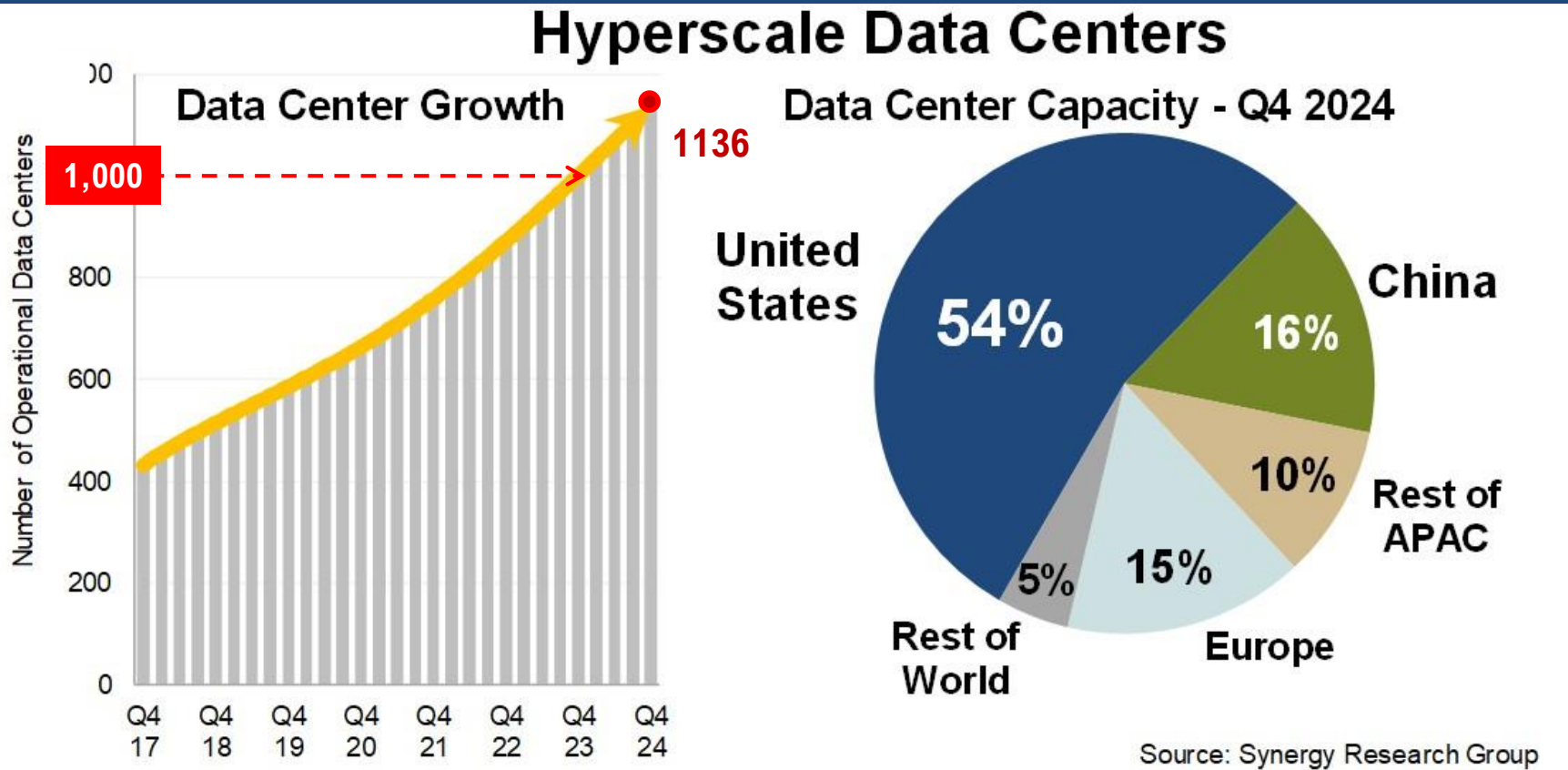
BY ARCHITECTURE

- ❖ Hyperscale Data Centers
- ❖ Edge Data Centers
- ❖ Micro Data Centers
- ❖ Modular/Container Data Centers

KEY COMPANIES PROFILED



超大型資料中心成長



The number of large data centers operated increased to 1,136 at the end of 2024, having doubled over the last five years. It has taken < 4 years for the total capacity of operational hyperscale data centers to double, as the average capacity of newly opened facilities continues to climb. The US still accounts for well over half of total worldwide capacity, with Europe and China each accounting for about 1/3 of the balance. Looking ahead, Synergy forecasts that it will take < 4 years for total hyperscale data center capacity to double once again.

超大型資料中心市場

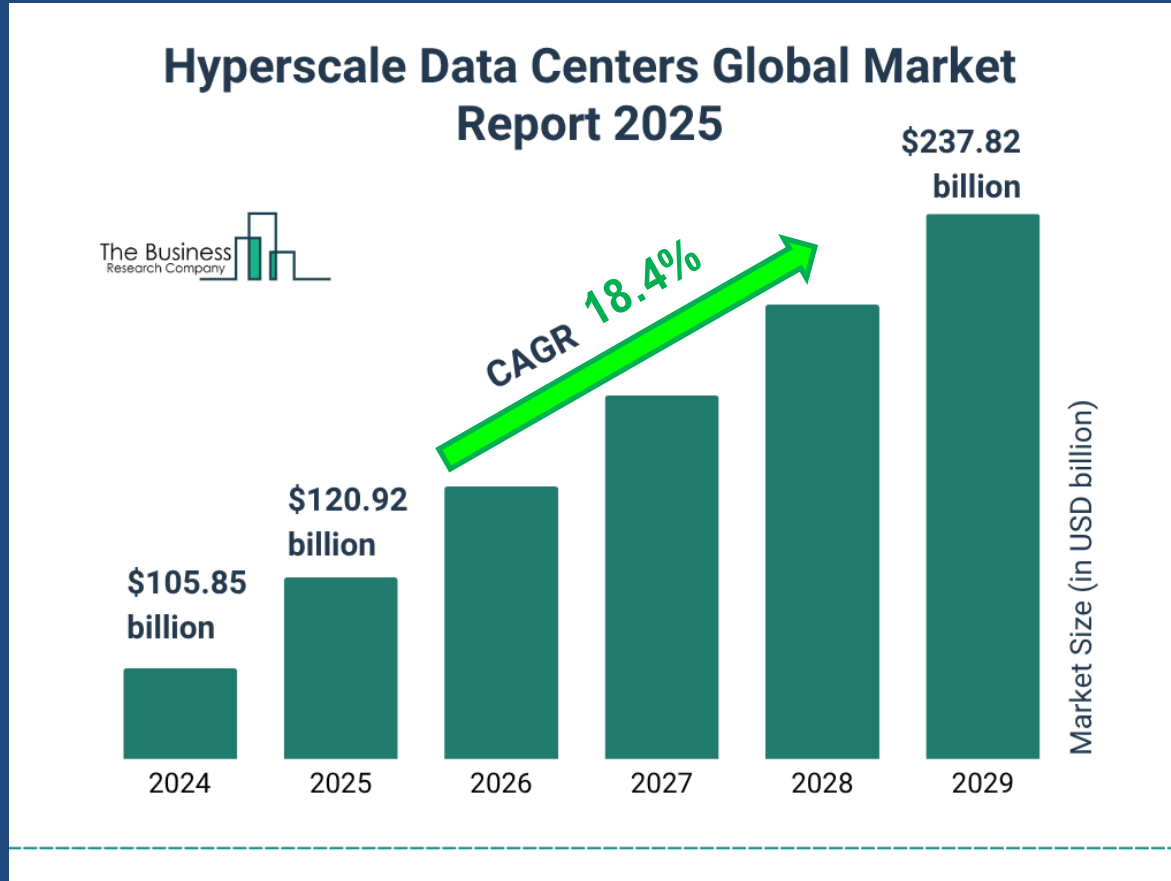
成長動能:

過去:

- ✓ Cloud computing growth
- ✓ Internet services
- ✓ Data explosion
- ✓ Energy efficiency

未來:

- ✓ Edge computing
- ✓ Cloud services
- ✓ AI and machine learning



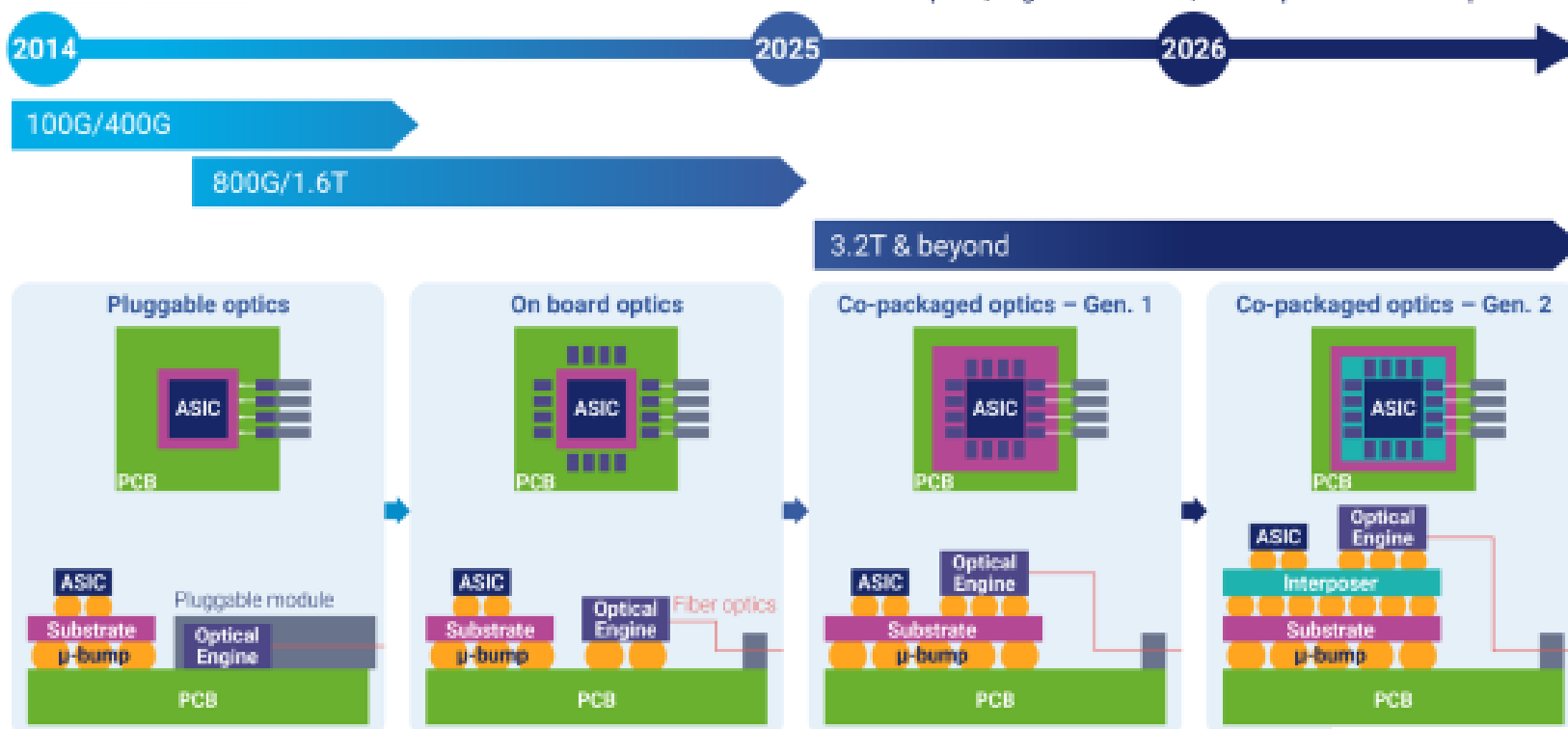
Source: The Business Research Company, 2025

CPO 解决方案

Shorter Electrical Path / Higher Bandwidth / Lower Power Consumption

Key Trend of Optical Transceiver Packaging in High-End Data Center

More advanced package (higher complexity), shorter electrical path, high bandwidth, lower power consumption



Source: [IDTechEx](#) Research

> 10 pJ/b

1 ~ 10 pJ/b

< 1 pJ/b

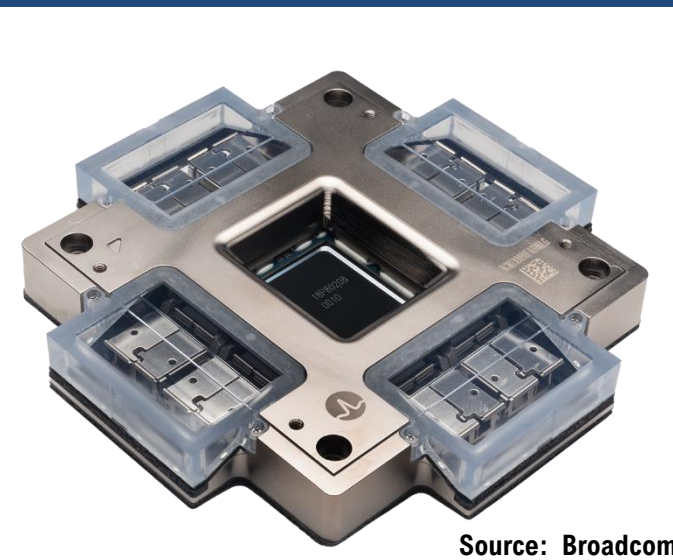
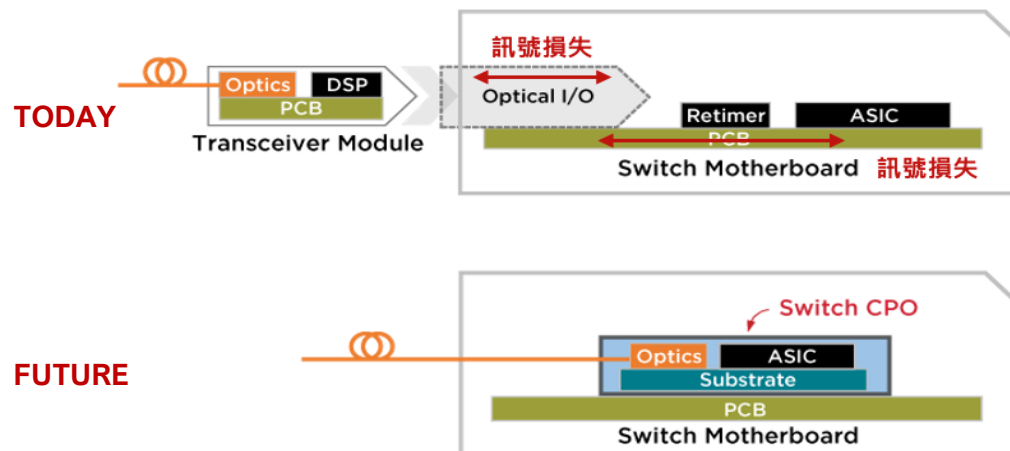
CPO – Co-Packaged Optics

共同結構光學

Broadcom 方案



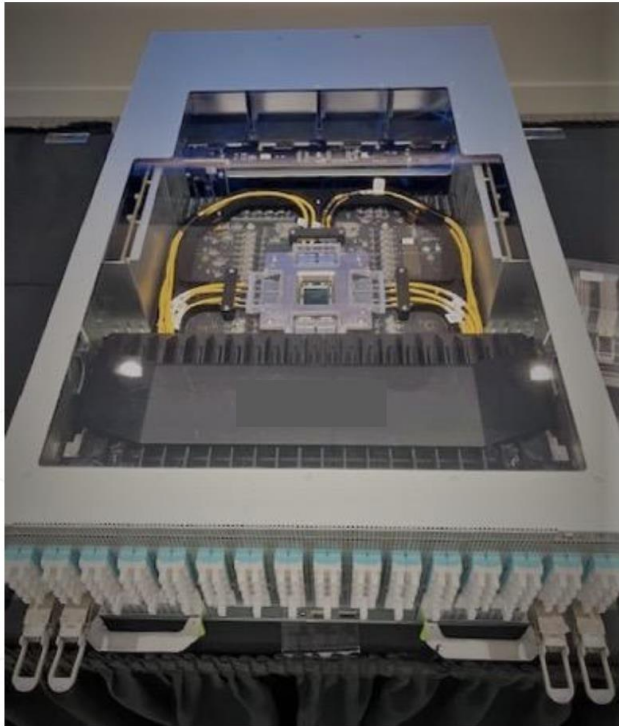
CPO 設計: 將IC元件及光學元件集成在單一封裝基板



Source: Broadcom

51.2T CPO Switch

TH5-Bailly: Second Generation System

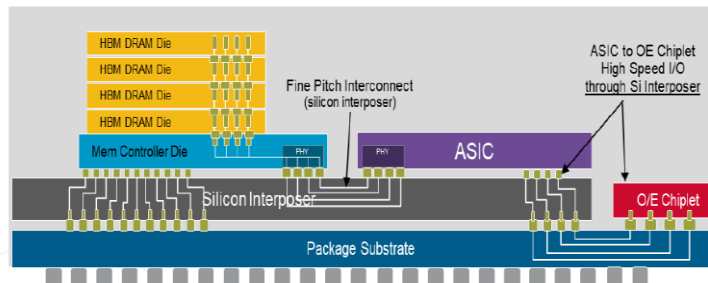


Source: Broadcom

Product Features:

- 51.2T Ethernet Switch
- All Optical CPO connectivity
- Eight 6.4T optical engines (64x100Gbps FR4 connectivity)
- Optical engine is a PIC bonded to a CMOS EIC
- Each optical engine has ~ 1000 optical components

Broadcom 領先業界推出 51.2T 全光交換機

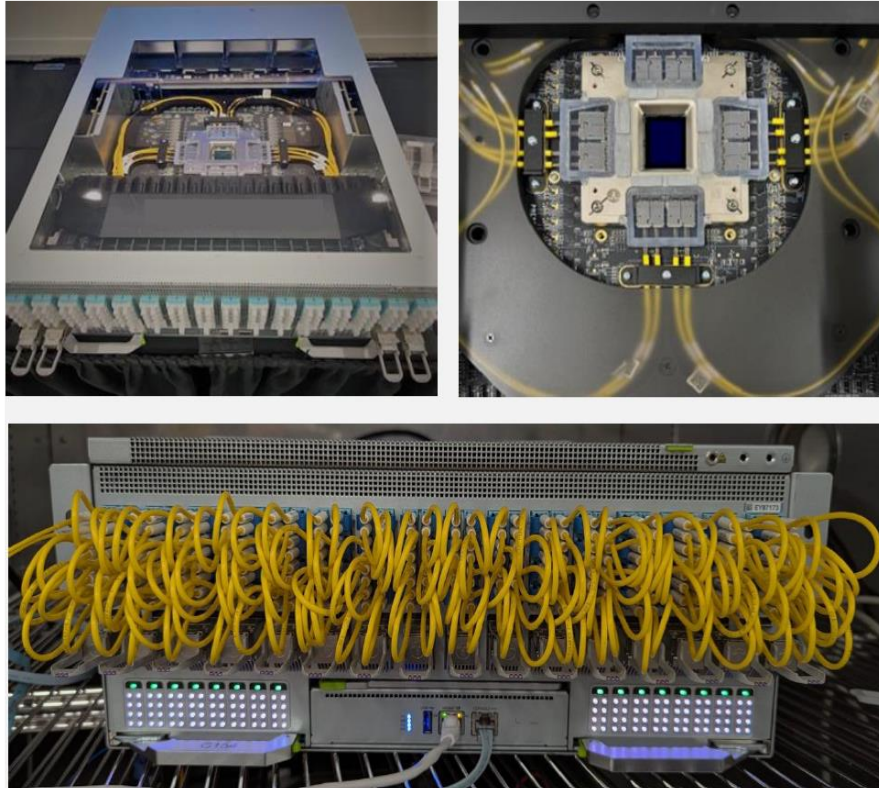


AI scaling with 2.5D multi-die packaging

CoWoS Package with Si Interposer, O/E chiplets and HBM

CPO with 6.4Tbps I/O BW per optical Engine

CPO Technology



Broadcom TH5-Bailly CPO

CPO 可靠性驗證系統

Rack Level



Chip Level

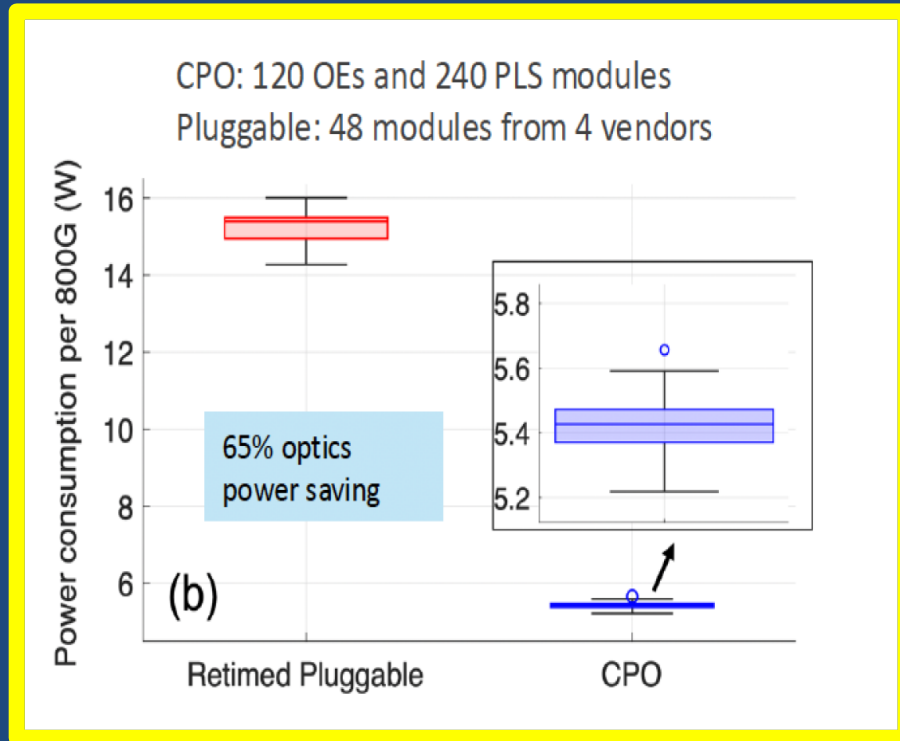


Complete systems are assembled & installed in a rack, and monitored for change in traffic performance while operating above maximum specified ambient temperature.

CPO chip level reliability testing at even further elevated temperatures than rack units, increasing the acceleration factor of the test.

TH5-Bailly CPO Performance

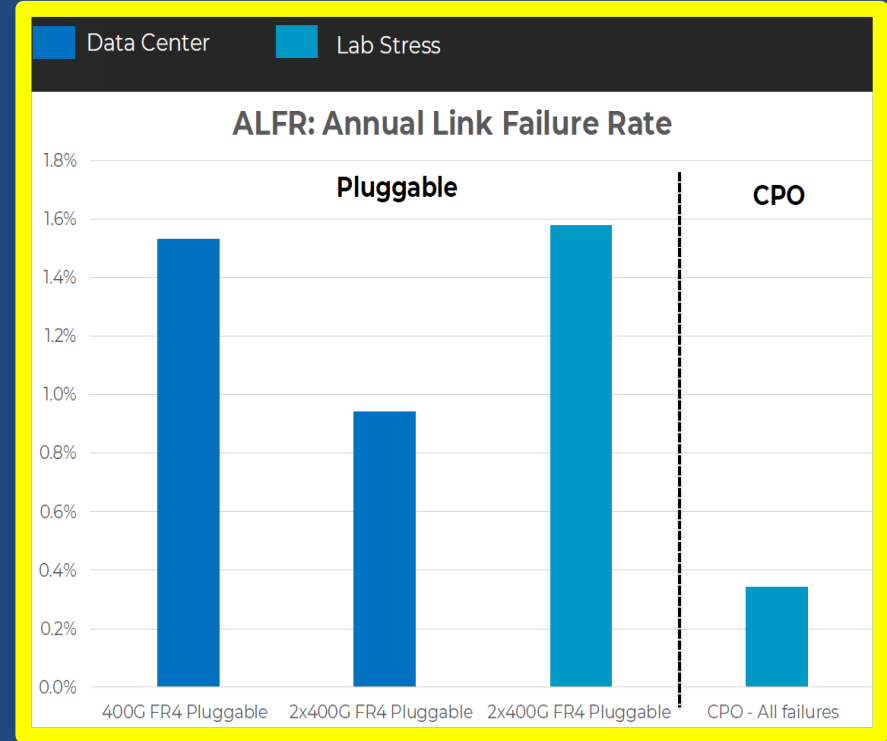
Power:



✓ **65%** CPO power savings vs. retimed optics

✓ **35%** CPO power savings vs. LPO for 100G/lane

ALFR:

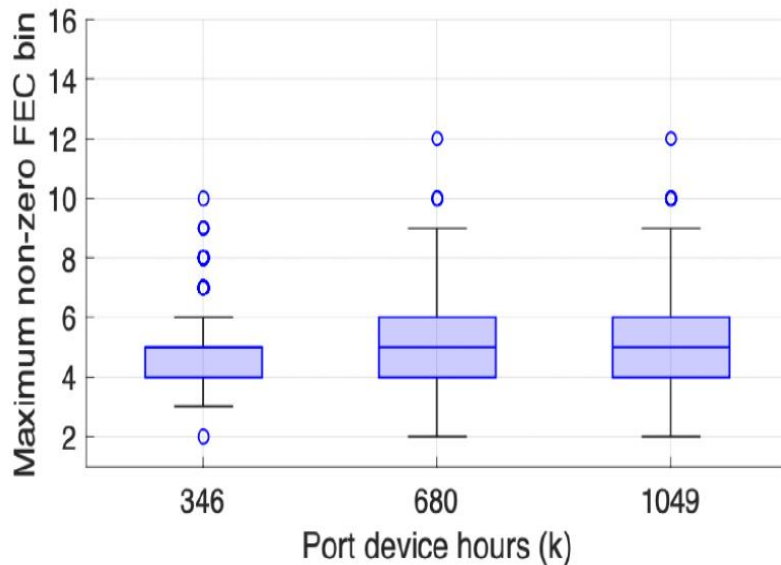


✓ **5X** CPO serviceable failures improvement over pluggables

TH5-Bailly CPO Performance

FEC:

FEC statistics over time for the first 1M device hours



TH5-Bailly CPO HTOL Stress Hours

CPO HTOL Systems	Tot. QTY	Sept 29, 2025
Rack Unit systems	27	52,721 hrs
Mezz Card systems	43	76,204 hrs
Total HTOL Stress	70	128,925 hrs

✓ **1M**

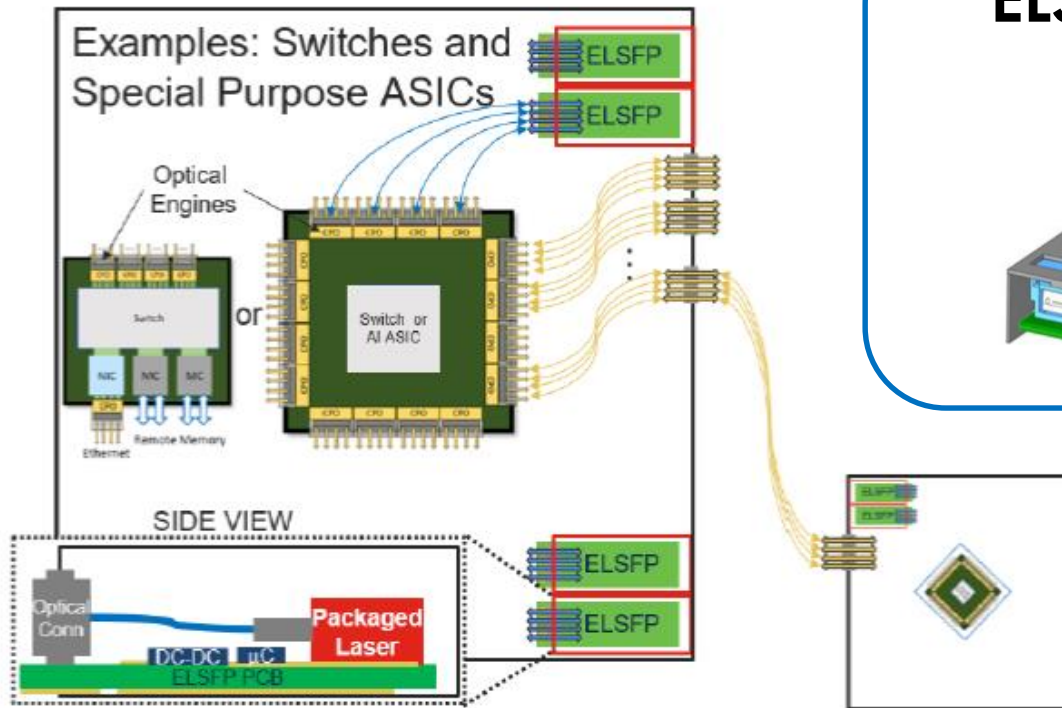
No Link Flaps observed in first 1M CPO device hours

✓ **8.5M**

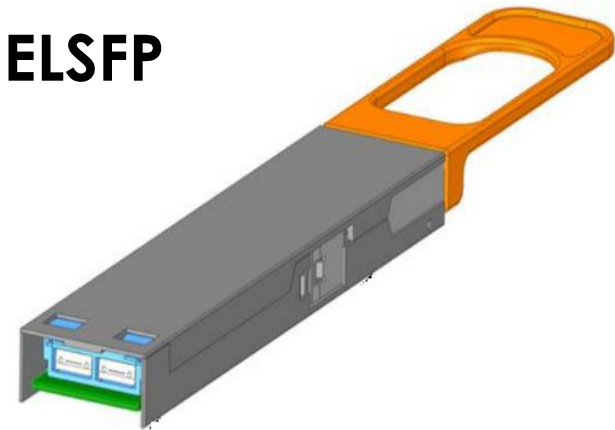
800G Equivalent devices hours of HTOL stress

CPO ELSFP

Objective: Ensure a safe method to field replace lasers in the event of failure. Flexibility has been added to the form factor to enable a range of applications from massive scale data center switches to special purpose optically enabled ASICs.



ELSFP

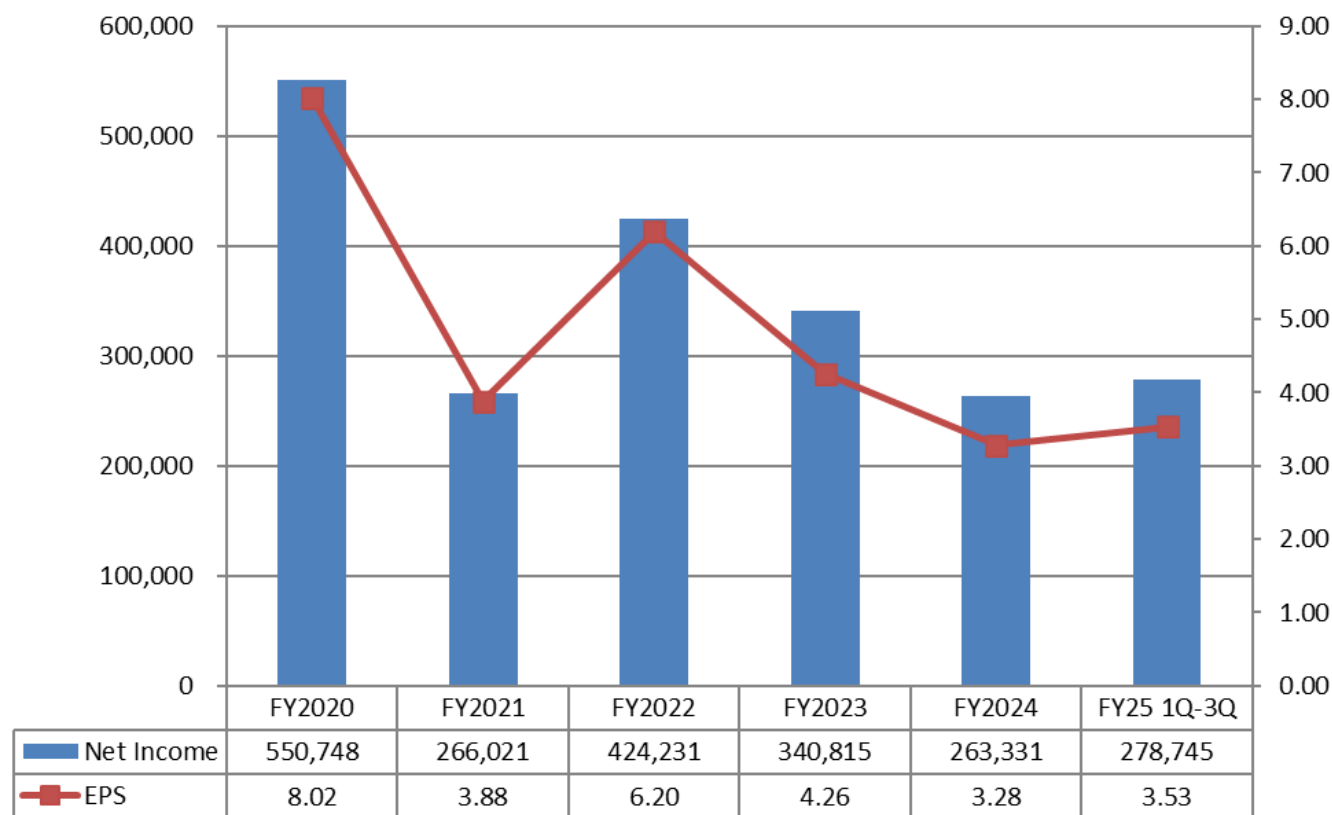


核心設計優勢:

- ✓ 熱管理
- ✓ 可維護性
- ✓ 安全性
- ✓ 靈活性

財務表現

稅後淨利 (新台幣仟元) 及每股獲利 (新台幣元)



THANK YOU